

sunSearchPoint

1 Executive Summary

This module drives the spacecraft's safe-mode startup in two phases: a scripted **sun search** followed by closed-loop **sun pointing**. On reset it begins a sequence of constant-rate body rotations that sweep the body-fixed coarse sun sensors (CSS) through different orientations. Once the sun is acquired — signalled by the CSS observation count reaching a configurable threshold — the module transitions to sun pointing, producing attitude tracking errors as Modified Rodrigues Parameters (MRPs) and body-rate tracking errors that align a commanded body axis with the sun heading. An optional constant spin rate about the sun heading axis can be specified. If the sun direction is not available during pointing, the attitude error is zeroed and a configurable body-fixed fallback rate is commanded instead.

The phase transition is **one-way**:

- The first rotation always runs to completion.
- After it, each update transitions to pointing when `sizeOfObservations` is at least `observationThreshold`.
- If the full rotation sequence elapses without the sun being acquired, the module transitions to pointing regardless (forced).
- Once pointing, the module never returns to search (re-armed only by `reset()`).

The output should be paired with a control module such as `rateControl` / `mrpFeedback`.

2 Message Connection Descriptions

The following table lists all the module input and output messages.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
<code>sunDirectionInMsg</code>	<code>NavAttMsgF32Payload</code>	input message containing the sun direction vector s in body frame coordinates
<code>rateInMsg</code>	<code>NavAttMsgF32Payload</code>	input message containing the inertial body angular velocity $\omega_{B/N}$
<code>filterResidualsInMsg</code>	<code>FilterResidualsMsgF32Payload</code>	input message whose <code>sizeOfObservations</code> field carries the number of valid CSS observations, used to detect sun acquisition
<code>attGuidanceOutMsg</code>	<code>AttGuidMsgF32Payload</code>	output message of attitude tracking errors and reference frame states
<code>sunSearchPointFaultOutMsg</code>	<code>SunSearchPointFaultMsgPayload</code>	output message whose <code>faultDetected</code> flag latches true if the search sequence elapses without acquiring the sun (the forced transition to pointing); re-armed only by <code>reset()</code>

3 Module Parameters

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description
sHatBdyCmd (required)	Eigen::Vector3f	[-]	zero	Commanded body-fixed unit direction vector $\hat{\mathbf{s}}_c$ to align with the sun (norm must be within $1\mathbf{e-3}$ of 1.0; renormalized on storage)
sunAxisSpinRate	float	[rad/s]	0	Desired constant spin rate $\dot{\theta}$ about the sun heading vector
omega_RN_B	Eigen::Vector3f	[rad/s]	zero	Body-fixed fallback rate commanded in the pointing phase when no sun direction is available
observationThreshold	int	[-]	4	CSS observation count at or above which the sun is considered acquired
rotations	RotationProperties[4]		1 s, 0 rad/s, b1	Sun-search rotation sequence; each entry has a duration (> 0), a signed body rate, and an axis. Set via <code>setRotation(index, rotation)</code> .

4 Module Assumptions and Limitations

- The input sun direction vector $\mathbf{s}$ can have any length. Only the exact zero vector is treated as sun-not-available; any non-zero magnitude is accepted.
- The commanded body-relative unit direction vector $\hat{\mathbf{s}}_c$ is fixed relative to the body.
- The sun-pointing condition is under-determined (2 DOF): the rotation about $\mathbf{s}$ is arbitrary.
- Each rotation's duration must be finite and strictly positive; the commanded rate must be finite (any sign). Invalid sequences are rejected when the configuration is installed.

5 Initialization

The module is configured by:

```

module = sunSearchPointF32.SunSearchPoint()
module.modelTag = "sunSearchPoint"
module.sHatBdyCmd = [0.0, 0.0, 1.0]
module.sunAxisSpinRate = 0.0
module.omega_RN_B = [0.0, 0.0, 0.0]
module.observationThreshold = 4

rotation = sunSearchPointF32.RotationProperties()
rotation.rotationDuration = 30.0
rotation.rotationRate = 0.1
rotation.rotationAxis = sunSearchPointF32.RotationAxis_b1Hat_B
module.setRotation(0, rotation)
# ... repeat for indices 1..3

```

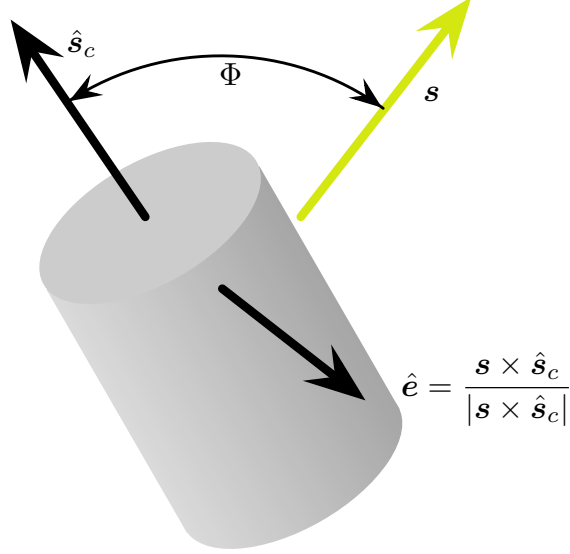


Figure 1: Body Vector Illustrations

6 Detailed Module Description

6.1 Phase State Machine

On the first update after `reset()` the module latches the current time as the search start and enters the SEARCH phase. Let t_e be the elapsed time since the start and T_k the cumulative end time of rotation k . The transition to POINT occurs when

$$(t_e \geq T_0 \text{ and } \text{sizeofObservations} \geq \text{observationThreshold}) \quad \text{or} \quad t_e \geq T_{N-1}$$

where N is the number of rotations. POINT is terminal. When the transition is *forced* by the second condition alone — the full sequence elapses ($t_e \geq T_{N-1}$) without the observation count ever reaching the threshold — the search has failed to acquire the sun. In that case the module latches a fault: `sunSearchPointFaultOutMsg.faultDetected` is set `true` and stays `true` for the rest of the run (even if the sun is later seen), cleared only by `reset()`.

6.2 Search Phase

For each active rotation the module outputs a constant reference angular velocity $\omega_{R/N}$ along the corresponding body axis and zero attitude error $\sigma_{B/R} = \mathbf{0}$. The active rotation is the first whose cumulative end time has not yet elapsed. The body-rate error is

$$\omega_{B/R} = \omega_{B/N} - \omega_{R/N}$$

6.3 Pointing Phase — Good Sun Direction Vector

The principal rotation vector and angle aligning $\hat{s}_c$ with the sun heading $\mathbf{s}$ are

$$\hat{\mathbf{e}} = \frac{\mathbf{s} \times \hat{\mathbf{s}}_c}{|\mathbf{s} \times \hat{\mathbf{s}}_c|} \quad \Phi = \arccos\left(\frac{\mathbf{s} \cdot \hat{\mathbf{s}}_c}{|\mathbf{s}|}\right)$$

The attitude error MRP and reference rate are

$$\sigma_{B/R} = \tan\left(\frac{\Phi}{4}\right) \hat{\mathbf{e}} \quad {}^B\omega_{R/N} = {}^B\hat{\mathbf{s}}\dot{\theta}$$

with rate tracking error $\omega_{B/R} = \omega_{B/N} - \omega_{R/N}$ and zero reference acceleration $\dot{\omega}_{R/N} = \mathbf{0}$.

6.4 Pointing Phase — No Sun Direction Vector

If $\mathbf{s}$ is the zero vector, the attitude error is set to zero and the configured fallback rate is commanded: $\sigma_{B/R} = \mathbf{0}$, $\omega_{R/N} = \text{omega_RN_B}$. This is the branch reached when the search sequence elapses without acquiring the sun, keeping the vehicle slewing rather than at rest. The branch is evaluated every update, so re-acquiring the sun resumes pointing immediately.

6.5 Collinear Commanded and Sun Heading Vectors

When $\mathbf{s} \approx \hat{\mathbf{s}}_c$ the cross product vanishes; `stableNormalized` returns the zero vector and $\tan(\Phi/4) \rightarrow 0$, so $\sigma_{B/R} = \mathbf{0}$ without a special case. When $\mathbf{s} \approx -\hat{\mathbf{s}}_c$ (180°), a fallback eigen-axis orthogonal to $\hat{\mathbf{s}}_c$ is used: $\hat{\mathbf{e}}_{180} = \text{sHatBdyCmd.unitOrthogonal}()$.

7 Test Description

The module is verified through regression tests against an independently coded reference implementation of the pointing logic, property-based tests (MRP norm bounds, the rate identity $\omega_{B/R} = \omega_{B/N} - \omega_{R/N}$, finiteness), edge-case tests (collinear and 180° vectors, zero-vector sun-not-visible, the $\hat{\mathbf{e}}_{180}$ fallback), and search/state-machine tests covering active-rotation selection, the first-rotation commit window, the observation-threshold transition, the forced transition after the full sequence, the terminal pointing phase, and the zero-sun fallback rate. The module adapter is exercised in Python for input/output wiring, the search-to-point transition, and the `reset()` link checks.